Rural Vaccine Cold Chain

Section one introduces the cold chain problem facing pharmaceutical distributors in rural districts. Vaccines and insulin must stay between two and eight degrees Celsius, yet many primary health centres lose grid power for several hours a day.

We surveyed published reports from state health departments and found that temperature excursions are rarely logged, which makes it hard to know how much medicine is actually spoiled.

Solar powered refrigerators with phase change batteries can keep vaccines cold for up to five days without electricity. Each unit costs about three thousand dollars, but it removes the need for diesel generators and cuts spoilage to almost zero. We recommend that every district deploy these units at the ten health centres with the worst power supply first.

A low cost temperature logger with a cellular connection should be attached to every unit, so that district officials receive an alert whenever the temperature leaves the safe range.